

Ahmed Hassan Khawaja

AI / Software Engineer

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PROFESSIONAL SUMMARY

Software engineering student at UNSW, passionate about programming and artificial intelligence. Enjoys building practical tools, from automation pipelines and AI-powered applications to full-stack web products, that solve real problems for real users. Comfortable across Python, JavaScript, and cloud engineering, and always eager to learn new technologies and ship high-quality software.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C, C++

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), prompt engineering, AI agents, vector databases, data extraction & normalisation

Cloud & DevOps: Amazon Web Services (AWS): ECS, EC2, Lambda; Docker, Git, CI/CD

Frameworks & Libraries: React, Next.js, Node.js, Express, Playwright, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, vector databases

Concepts: REST API design, web scraping & automation, data pipelines, event-driven systems, Agile

PROFESSIONAL EXPERIENCE

AI Software Engineer Intern

Effi Technologies Pty Ltd

May 2026 to July 2026

Sydney, NSW

- Built an automated cashback ingestion pipeline in Python and Playwright that scrapes offers from **50+ banking partners** and loads them into the company platform daily, saving **~15 hours** of manual effort per week.
- Deployed the pipeline on AWS (ECS, EC2, Lambda) with Docker, cutting end-to-end processing time from several hours to **under 20 minutes**.
- Engineered resilient scraping and parsing logic handling **10,000+ records per day**, lifting data accuracy above **99%** and reducing failed runs by **~90%**.
- Integrated AI/LLM-based extraction to normalise unstructured offer data across bank formats, cutting manual correction by **~70%**.

Software Engineer Intern

Nextbridge Ltd.

Dec 2025 to Feb 2026

- Built and deployed scalable MERN-stack web applications in production, delivering features across **3+ client workflows**.
- Developed RESTful APIs with Node.js and Express, and implemented secure event-driven payment processing via Stripe webhooks with near-zero failed transactions.
- Implemented real-time client and server communication with WebSockets to power live updates and reduce user-facing latency.

Founder & Full-Stack Developer

Sydney Design & Development

2024 to Present

Sydney, NSW

- Founded a web studio delivering full-stack applications for Sydney-based businesses, launching a live marketplace platform (sydneydesigndevelopment.com.au).
- Architected and managed database systems supporting **1,500+ products**, with secure payment integration and an admin dashboard.

PROJECTS

Teapots Invoicing, AI Document Automation (SENG2021)

2026

Awarded UNSW Faculty of Engineering, Top Teams Recognition

React, Node.js, AWS

- Built a web tool that automates creation of invoices, despatch advices, and order documents, with AI-assisted generation and validation of document data.
- Deployed the application on AWS and published it live at teapotinvoicing.app.
- Recognised by the UNSW Faculty of Engineering as a Top Team (Term 1 2026) for exceptional technical work, adaptability, and user-centred design.
- Delivered as a 4-person team using Agile, adapting to changing requirements and iterating on user feedback.

ServiceUNSW Digital Identity, VIP CHALLENGE (Engineering)

2026

University of New South Wales

NFC, Backend, Security

- Designing an NFC-enabled digital identity system for secure campus access and authentication as part of UNSW's Vertically Integrated Projects (VIP) program.
- Architecting backend identity and access-control infrastructure with data-security controls and real-time verification workflows.

Farm Fresh, Inventory Management System

2025

Full-stack web app for a produce farm in the United States

React, Node.js, MongoDB

- Built a full-stack platform that helps a US farm track and manage its produce inventory in real time, handling **100+ produce records**.
- Developed React UI components and REST APIs on Node.js, Express, and MongoDB, added role-based access control, and deployed the system to production on Vercel and Render.

EDUCATION

University of New South Wales (UNSW)

Expected 2028

Bachelor of Engineering (Honours), Software · Distinction (WAM)

Sydney, NSW

Ranked Australia's #1 university and top 20 globally (QS World University Rankings 2026).

Relevant coursework:

- Database Systems (COMP3311): **88**
- Programming Fundamentals: **87**
- Requirements & Design Workshop (SENG2021): **85**
- Data Structures & Algorithms: **84**
- Computer Networks & Applications (COMP3331): **80**

Aitchison College

A-Levels: 1A* 3A (ATAR equivalent 99.00).

LEADERSHIP & RECOGNITION

- **Top Teams Recognition, UNSW Faculty of Engineering:** awarded (Term 1 2026) for the Teapots Invoicing project.
- **Marketing Executive, Arc @ UNSW Student Club:** led a team of 10+ designers producing brand assets for club events and campaigns.
- **General Secretary, Computer Society (Aitchison College):** organised hackathons and a national STEM Olympiad with 1,000+ delegates.
- **Course Contributor, Z-Notes:** authored IGCSE Computer Science revision content used by 6M+ students.
- **FinTech Judge, Aitchison Business Concept:** selected to judge fintech startup pitches at the college's flagship competition.